CHAPSim2: Solver Basics and Workflow

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Outline

CHAPSim2: What is it?

CHAPSim2: Workflow

CHAPSim2: Code

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CHAPSim2: Workflow

CHAPSim2: Code

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CHAPSim History

From Research Code to Community Capability

CHAPSim
CHANNEL AND PIPE FLOW SIMULATION

CHAPSim is a high-fidelity Direct Numerical Simulation (DNS) solver for incompressible turbulent flow and heat transfer in canonical configurations such as channel and pipe flows.

The current actively developed version is CHAPSim2, which provides higher-order numerical accuracy and improved parallel performance for modern HPC systems.

DNS • HIGH-ORDER • HPC

2007–2012
Origin

PhD Development
- Developed by Dr Mehdi Seddighi during his PhD (from 2007)
- Focus on isothermal, incompressible, unsteady turbulent channel flow
- Implemented in Fortran 77
- Based on numerical approaches from Orlandi (2000)

2013–2015
Extension & Modernisation

Thermo-fluid Extension
- Extended by Dr Wei Wang
- Added thermo-fluid capability
- Redeveloped in modern Fortran
- Enhanced numerical capability and stability

2020
First Open-Source Release

CHAPSim1.0
- First open-source release within CCP-NTH
- Second-order spatial discretisation
- One-dimensional parallelisation
- Retained for reference only
- No longer actively supported

2020–Present
Community Development

Community Code
- Evolved into an open-source community capability
- Development led within CCP-NTH
- Contributions from STFC Daresbury Laboratory and the University of Sheffield

2026
Current Production Release

CHAPSim2 (v2.1.0)
- Higher-order accuracy (up to sixth order)
- Multi-dimensional parallelisation (2-D decomposition)
- Improved scalability for modern HPC systems
- Latest production release: **v2.1.0** (30 January 2026)

Purpose
Provide a high-fidelity, reliable and scalable DNS platform for incompressible turbulent flows and heat transfer in channel, pipe and related configurations.

Open-Source & Community Driven
CHAPSim2 is released as open-source software under the **BSD 3-Clause License.**

BSD 3-Clause License

CHAPSim2: What is it for?

Core purpose

- High-fidelity DNS solver
- Turbulent flow and heat transfer
- Designed for canonical wall-bounded flows:
  - channel, pipe, annular

Supported physics

- Isothermal flows
- Thermal & variable-property flows
- Magnetohydrodynamics (MHD) via imposed-field Lorentz-force coupling

Typical applications

- Supercritical fluids
- Liquid metals
- Heated/cooled turbulence
- MHD flows

CHAPSim2 is designed for **physics-resolved DNS** in controlled geometries for research and model development.

CHAPSim2: Physics and Numerical Core

Geometry & flexibility

- Both Cartesian and cylindrical coordinates
- Applicable to wide fluid regimes:
  - classical fluids
  - supercritical fluids
  - liquid metals

Numerical formulation

- Fully staggered grid
- Conservative variables: mass flux ρu_i and enthalpy-related conservative variable ρh

Spatial discretisation

- 2nd and 4th order central finite differences (explicit)
- 2nd, 4th, and 6th order compact (implicit) schemes

Temporal discretisation

- 3rd order Runge-Kutta (explicit)

Grid capability

- 1-D grid stretching in wall-normal direction
- Supports periodic, wall-bounded, and inlet/outlet-type boundary conditions

CHAPSim2 combines high-order accuracy with robust conservation for DNS of complex thermo-fluid systems.

CHAPSim2: HPC and Usability

HPC performance

- MPI parallelisation
- 2D pencil decomposition
- Based on 2decomp&FFT library

FFT capability

- 2D and 3D FFT solvers
- Efficient 3-D Poisson solver
- Scalable spectral operations

User interface

- Fortran solver core
- Python interface:
  - input setup
  - visualisation

Designed for large-scale DNS while retaining practical setup, monitoring and post-processing workflows.

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Governing Equations in CHAPSim2 (Cylindrical Formulation)

Primary and conservative variables

$$q_x = u_x, \quad q_r = r u_r, \quad q_\theta = u_\theta$$

$$g_x = \rho u_x, \quad g_r = r \rho u_r, \quad g_\theta = \rho u_\theta$$

Velocity divergence

$$\Psi = \frac{\partial q_x}{\partial x} + \frac{1}{r}\frac{\partial q_r}{\partial r} + \frac{1}{r}\frac{\partial q_\theta}{\partial \theta}$$

Continuity equation

$$\frac{\partial \rho}{\partial t} + \frac{\partial g_x}{\partial x} + \frac{1}{r}\frac{\partial g_r}{\partial r} + \frac{1}{r}\frac{\partial g_\theta}{\partial \theta} = 0$$

Pressure Poisson equation

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 \phi}{\partial \theta^2} = \Psi$$

Key points

CHAPSim2 employs a **conservative formulation** to improve stability and discrete consistency. The radial momentum variable is defined in a weighted form ($q_r = r u_r$), which simplifies the governing equations and helps regularise the discrete formulation near the centreline, but still

Momentum Equation: x-direction

Conservative x-momentum equation

$$\frac{\partial g_x}{\partial t} + \frac{\partial (g_x q_x)}{\partial x} + \frac{1}{r}\frac{\partial (g_r q_x)}{\partial r} + \frac{1}{r}\frac{\partial (g_\theta q_x)}{\partial \theta} = -\frac{\partial p}{\partial x}$$
$$+ \frac{\partial}{\partial x}\left[2\mu\left(\frac{\partial q_x}{\partial x} - \frac{1}{3}\Psi\right)\right]$$
$$+ \frac{1}{r}\frac{\partial}{\partial r}\left[\mu\left(\frac{\partial q_r}{\partial x} + r\frac{\partial q_x}{\partial r}\right)\right]$$
$$+ \frac{1}{r}\frac{\partial}{\partial \theta}\left[\mu\left(\frac{\partial q_\theta}{\partial x} + \frac{1}{r}\frac{\partial q_x}{\partial \theta}\right)\right]$$

Geometry terms

The highlighted r terms come from the cylindrical-coordinate formulation.

Momentum Equation: r-direction

Conservative r-momentum equation

$$
\begin{aligned}
\frac{\partial g_r}{\partial t} + \frac{\partial (g_x q_r)}{\partial x} + \frac{\partial (g_r q_r / r)}{\partial r} + \frac{\partial (g_\theta q_r / r)}{\partial \theta} - \varepsilon\, g_\theta q_\theta = -r \frac{\partial p}{\partial r} \\
+ \frac{\partial}{\partial x}\left[\mu\left(\frac{\partial q_r}{\partial x} + r\frac{\partial q_x}{\partial r}\right)\right] \\
+ \frac{\partial}{\partial r}\left[r\, 2\mu\left(\frac{\partial (q_r/r)}{\partial r} - \frac{1}{3}\Psi\right)\right] \\
+ \frac{\partial}{\partial \theta}\left[\mu\left(\frac{\partial q_\theta}{\partial r} + \frac{1}{r}\frac{\partial (q_r/r)}{\partial \theta} - \varepsilon\frac{q_\theta}{r}\right)\right] \\
- \varepsilon\left[2\mu\left(\frac{1}{r}\frac{\partial q_\theta}{\partial \theta} + \frac{q_r/r}{r} - \frac{1}{3}\Psi\right)\right]
\end{aligned}
$$

The r-momentum equation contains both metric terms (r) and curvature terms (ε), leading to strong coupling between velocity components and additional complexity compared with Cartesian formulations.

Momentum Equation: θ-direction

Conservative θ-momentum equation

$$
\begin{aligned}
\frac{\partial g_\theta}{\partial t} + \frac{\partial (g_x q_\theta)}{\partial x} + \frac{1}{r^2}\frac{\partial (r g_r q_\theta)}{\partial r} + \frac{1}{r}\frac{\partial (g_\theta q_\theta)}{\partial \theta} &= -\frac{1}{r}\frac{\partial p}{\partial \theta} \\
&+ \frac{\partial}{\partial x}\left[\mu\left(\frac{\partial q_\theta}{\partial x} + \frac{1}{r}\frac{\partial q_x}{\partial \theta}\right)\right] \\
&+ \frac{1}{r}\frac{\partial}{\partial r}\left[\mu\left(r\frac{\partial q_\theta}{\partial r} + \frac{1}{r}\frac{\partial q_r}{\partial \theta} - \varepsilon q_\theta\right)\right] \\
&+ \frac{1}{r^2}\frac{\partial}{\partial \theta}\left[2\mu\left(\frac{\partial q_\theta}{\partial \theta} + \varepsilon\frac{q_r}{r} - r\frac{1}{3}\Psi\right)\right] \\
&+ \varepsilon\frac{1}{r}\left[\mu\left(\frac{\partial q_\theta}{\partial r} + \frac{1}{r^2}\frac{\partial q_r}{\partial \theta} - \frac{q_\theta}{r}\right)\right]
\end{aligned}
$$

Cylindrical geometry introduces metric (r) and curvature (ε) effects, resulting in additional source and coupling terms that do not exist in channel (Cartesian) flows.

Energy Equation (Conservative Form)

Energy equation in cylindrical coordinates

$$\frac{\partial(\rho h)}{\partial t} + \frac{\partial(g_x h)}{\partial x} + \frac{1}{r}\frac{\partial(g_r h)}{\partial r} + \frac{1}{r}\frac{\partial(g_\theta h)}{\partial \theta} =$$
$$\frac{\partial}{\partial x}\left(\kappa\frac{\partial T}{\partial x}\right) + \frac{1}{r}\frac{\partial}{\partial r}\left(r\kappa\frac{\partial T}{\partial r}\right) + \frac{1}{r}\frac{\partial}{\partial \theta}\left(\frac{1}{r}\kappa\frac{\partial T}{\partial \theta}\right)$$

Key points

The energy equation follows the same conservative structure as momentum, with additional geometric (r) effects in cylindrical coordinates.
Pressure-work and viscous-dissipation terms are omitted here under Low-Mach Number Approximation

Energy Equation: Closure and Workflow

Why enthalpy?

- Robust for variable-property flows
- Ensures strict energy conservation
- Handles strong temperature-dependent properties

Thermophysical closure

$$\rho,\ \mu,\ \kappa = f(T, p)$$

- Empirical correlations
- Look-up tables (e.g. supercritical fluids)

Solution workflow

$$\rho h, \rho \ \rightarrow\ h \ \rightarrow\ T \ \rightarrow\ \rho, \mu, \kappa$$

Coupling with momentum

- Properties depend on temperature
- Momentum energy strongly coupled
- Important in:
  - supercritical fluids
  - liquid metals

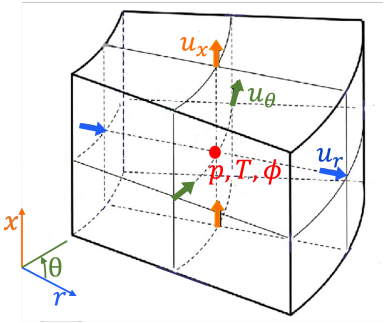
Fully Staggered Grid: Variable Arrangement

Grid layout (finite difference)

- Scalar variables stored at cell centres: $p,\ T,\ \phi$
- Velocity components stored at staggered locations:
  - u_x at x-faces
  - u_r at r-faces
  - u_θ at θ-faces

Discrete interpretation

- Staggering aligns variables with derivative locations
- Improves consistency of gradient and divergence operators



Cylindrical staggered grid arrangement

Fully Staggered Grid: Numerical Benefits

Pressure–velocity coupling

- Avoids pressure–velocity decoupling
- Eliminates checkerboard pressure modes
- Provides stable pressure Poisson formulation

Discrete consistency

- Gradient and divergence operators are naturally coupled
- Improves accuracy of continuity enforcement
- Reduces numerical oscillations

For CHAPSim2

- Consistent with conservative variables (g_i)
- Handles cylindrical metric terms (r) robustly
- Supports high-order finite-difference schemes

The staggered arrangement ensures stable coupling between pressure and velocity, and improves the accuracy of discrete differential operators in finite-difference DNS.

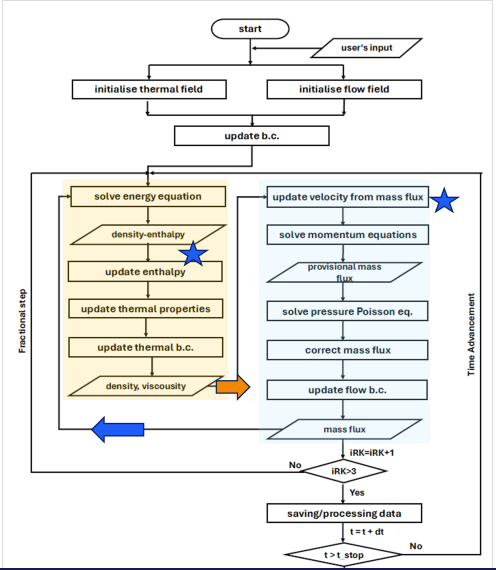
CHAPSim2 Workflow: Overview

Simulation workflow

- Initialise flow and thermal fields
- Apply boundary conditions
- Advance solution in time
- Solve coupled:
  - momentum + pressure
  - energy + properties
- Output and monitor results

Key idea

CHAPSim2 uses a fractional-step approach with strong coupling between flow and thermal fields.



CHAPSim2 Workflow: Time Advancement Loop

Flow solver loop

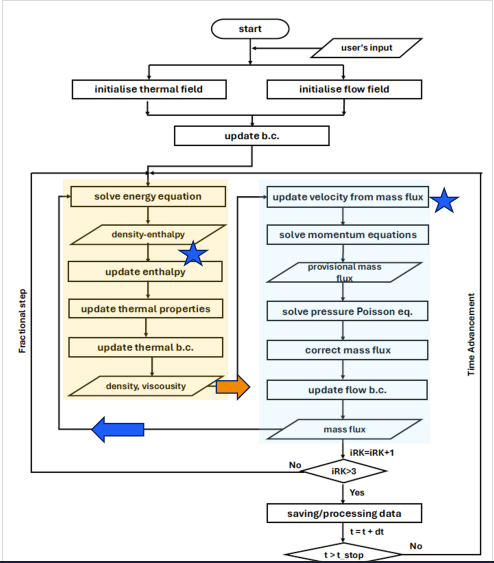
- Solve momentum equations
- Compute provisional mass flux
- Solve pressure Poisson equation
- Correct mass flux (continuity)

Thermal solver loop

- Solve energy equation
- Update enthalpy and temperature
- Update properties:

$$\rho, \ \mu, \ \kappa$$

Key message



Finite-Difference Schemes on Staggered Grids

Central finite difference

- Explicit, local stencil
- Simple and robust
- Commonly used for derivatives and interpolation between staggered locations

$$\left.\frac{\partial f}{\partial x}\right|_i \approx \frac{f_{i+1} - f_{i-1}}{2\Delta x}$$

$$f_{i+1/2} \approx \frac{f_i + f_{i+1}}{2}$$

Compact finite difference

- Implicit finite-difference stencil
- Higher spectral-like resolution
- Useful for high-accuracy DNS on smooth grids

$$\alpha f'_{i-1} + f'_i + \alpha f'_{i+1} = \frac{a}{2\Delta x}\left(f_{i+1} - f_{i-1}\right)$$

$$+\frac{b}{4\Delta x}\left(f_{i+2} - f_{i-2}\right)$$

On a staggered grid, derivative and interpolation operators must consistently transfer variables between cell-centred and face-centred locations.

Pipe Flow: Singularity at the Pole

Why the pole is special

- Cylindrical coordinates are singular at:

$$r = 0$$

- Terms such as

$$\frac{1}{r}, \qquad \frac{1}{r^2}$$

appear in the governing equations.

- The radial direction is not uniquely defined at the centreline.

The pole treatment affects:

- derivative operators
- interpolation
- pressure Poisson solve
- stability of pipe DNS

Numerical issue

Naively applying the same stencil at the pole can produce multi-valued velocity components and numerical inconsistency.

Axis Reconstruction in CHAPSim2

General requirement

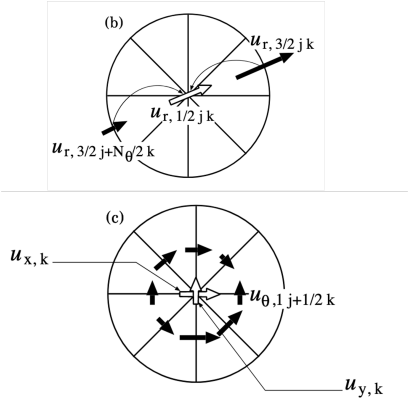
All physical variables must be:

- finite at $r = 0$
- single-valued (independent of θ)

Centreline Regularity: Fourier Behaviour at $r \to 0$

$$\phi(r,\theta) = \sum_{m=0}^{\infty} \phi_m(r)\, e^{im\theta}$$

$$\phi_m(r) \sim r^m \quad (r \to 0)$$



Using orthogonality relations of the trigonometric functions for variable reconstruction at pole.(Morinishi et al., 2004)

Pressure Poisson Equation in CHAPSim2

Why a Poisson equation is solved

- Momentum equations first give a provisional mass flux
- Pressure correction enforces continuity
- Conservative variables introduce a density-change source term

$$\nabla^2 \phi = \Psi$$

$$\Psi \sim \frac{1}{\Delta t}\left(\nabla \cdot \mathbf{g}^* + \frac{\partial \rho}{\partial t}\right)$$

Cylindrical form

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 \phi}{\partial \theta^2} = \Psi$$

Multiplying by r^2:

$$r^2\frac{\partial^2 \phi}{\partial x^2} + r\frac{\partial}{\partial r}\left(r\frac{\partial \phi}{\partial r}\right) + \frac{\partial^2 \phi}{\partial \theta^2} = r^2\Psi$$

The Poisson equation enforces mass conservation; in variable-density cylindrical flow, both $\partial\rho/\partial t$ and metric terms involving r must be treated consistently.

Poisson Solver Strategy: Cartesian vs Cylindrical

Cartesian cases

- Cartesian geometry has constant metric coefficients
- Homogeneous directions are efficiently handled spectrally
- CHAPSim2 uses a **3-D FFT-based Poisson solver**

Practical benefit

- Fast pressure correction
- Efficient parallelisation
- Suitable for high-order channel-flow DNS

Cylindrical cases

- Cylindrical metric terms introduce radial dependence
- FFT is used in homogeneous directions: x, θ
- Radial direction is solved using TDMA

Solver structure

2-D FFT in $(x, \theta) \rightarrow$ TDMA in $r \rightarrow$ inverse FFT

Important limitation

For pipe cases, the Poisson solver (2-D FFT + TDMA) is more constrained because the radial direction uses a second-order TDMA treatment.

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CHAPSim2 Code Structure

Core solver and build

- `src/`: main Fortran source code
- `build/`: build directory and Makefile
- `bin/`: compiled executable
- `obj/`: object files
- `build_chapsim.sh`: build script

External libraries

- `lib/2decomp-fft/`
  - domain decomposition
  - parallel FFT

User-facing folders

- `examples/`: tools for example simulation cases
- `tests/`: regression and smoke tests
- `prepost/`: pre/post-processing tools
- `docs/`: documentation

Pre/post-processing

- `prepost/autoinput/`: Python input-generation tools
- `prepost/useful_scripts/`: local/HPC helper scripts

Input File: `input_chapsim.ini`

Physical configuration

- Geometry: channel / pipe / annular
- Reynolds number or driving force
- Isothermal or thermal case
- Fluid properties (constant / variable)

Numerical setup

- Grid resolution and domain size
- Time step / CFL control
- Numerical schemes (order, options)

Boundary conditions

- Periodic / inlet–outlet setup
- Wall conditions (velocity, temperature, heat flux)
- Thermal buffer / ramp (if used)

Output & control

- Restart frequency
- Statistics sampling
- Visualisation output (XDMF/VTK)
- Diagnostics (mass, CFL, etc.)

Running a Simple Case

Typical workflow

```
# 1. Clone repository
git clone git@github.com:CCP-NTH/CHAPSim2.git
cd CHAPSim2

# 2. Build solver
./build_chapsim.sh

# 3. Enter a test case
cd tests/<case_name>

# 4. Edit input
vim input_chapsim.ini

# 5. Run
# On HPC systems, use the provided job script instead of direct mpirun.
mpirun -np 4 $PATH$/bin/CHAPSim
```

Minimal Working Case: What to Explore

Initial checks

- Does the code run successfully?
- Is CFL within stable range?
- Is mass conservation satisfied?
- Are output/restart files produced?

Basic diagnostics

- Monitor bulk velocity / flow rate
- Check no divergence or blow-up
- Inspect probe signals if available

Parameter exploration

- Reynolds number
- Grid resolution
- Time step (CFL)
- Output frequency
- Thermal options (if enabled)

Good practice

- Change one parameter at a time
- Compare against baseline case
- Keep runs short for testing

Bibliography

Morinishi, Y., Vasilyev, O. V., and Ogi, T. (2004). Fully conservative finite difference scheme in cylindrical coordinates for incompressible flow simulations. *Journal of Computational Physics*, 197(2):686–710.